Newton's Gravitational Constant from the Recognition Science Forcing Chain: Zero Free Parameters

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Abstract

We derive Newton's gravitational constant G from the Recognition Science (RS) forcing chain with zero free parameters. The derivation proceeds in two steps: (1) the Planck constant is $\hbar = \varphi^{-5}$ (in RS-native units), where the exponent $5 = 2^D - D$ counts the compact modes for $D = 3$ spatial dimensions; (2) the quantum-gravitational duality identity $G \cdot \hbar = 1$ is forced by the reciprocal symmetry $J(x) = J(x^{-1})$ of the cost functional. Together:

$$\boxed{G = \varphi^5 \approx 11.09 \quad \text{(RS-native units).}}$$

The recognition wavelength identity $(c^3 \lambda_{\text{rec}}^2)/(\hbar G) = 1/\pi$ provides an independent cross-check. The hierarchy between gravity and electromagnetism is explained by the φ-rung separation between the Planck and electronic scales.

1 Introduction

Newton's gravitational constant G is the least precisely known fundamental constant [2]:

$$G = 6.67430 \pm 0.00015 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}, \tag{1}$$

with relative uncertainty ~ 22 ppm, five orders of magnitude less precise than α (0.15 ppb). No relationship between G and other constants is known in conventional physics; G appears as a free parameter in Newtonian gravity and general relativity.

Recognition Science [7] derives G from the same forcing chain that determines α, $\hbar$, and c. The key insight is the *quantum-gravitational duality*: the J-cost functional's reciprocal symmetry $J(x) = J(x^{-1})$ forces $G \cdot \hbar = 1$ in RS-native units. Since $\hbar = \varphi^{-5}$ is already determined, $G = \varphi^5$ follows with zero free parameters.

2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness)**.** *The unique continuous solution to* (A1)–(A3) *is* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The *reciprocal symmetry* $J(x) = J(x^{-1})$ is the key structural property used in the derivation of G. Spatial dimension $D = 3$ is forced by the gap-45 synchronization $\mathrm{lcm}(8, 45) = 360$.

3 The RS Forcing Chain for Gravity

The derivation of G sits at the end of the RS forcing chain:

$$\text{RCL} \;\to\; J(x) \;\to\; \varphi \;\to\; 2^D = 8 \;\to\; c = 1 \;\to\; \hbar = \varphi^{-5} \;\to\; \boxed{G = \varphi^5.} \tag{2}$$

Step 1 (Cost functional). The RCL, normalization, and calibration uniquely force $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.
Step 2 (Self-similar ratio). The unique positive fixed point of the self-similar nesting equation is $\varphi = (1 + \sqrt{5})/2$.
Step 3 (8-tick epoch). The minimal ledger update period is $2^D = 8$ ticks for $D = 3$ spatial dimensions.
Step 4 (Coherence energy). The coherence energy is $E_{\text{coh}} = \varphi^{-5}$, with exponent $5 = 2^D - D$ counting the compact modes.
Step 5 (Planck constant). $\hbar = E_{\text{coh}} = \varphi^{-5}$ in RS-native units.
Step 6 (Gravitational constant). The quantum-gravitational duality $G \cdot \hbar = 1$ gives $G = 1/\hbar = \varphi^5$.

4 Derivation of G

4.1 The Quantum-Gravitational Duality

Lemma 4.1 (Reciprocal Symmetry of J)**.** *The J-cost functional satisfies $J(x) = J(x^{-1})$ for all $x > 0$.*

Proof. $J(x^{-1}) = \frac{1}{2}(x^{-1} + x) - 1 = \frac{1}{2}(x + x^{-1}) - 1 = J(x)$. □

Proposition 4.2 (Gravitational Constant — Structural Derivation)**.** *In RS-native units:*

$$\boxed{G = \varphi^5 \approx 11.09.} \tag{3}$$

Proof (structural argument). The cost functional's reciprocal symmetry $J(x) = J(x^{-1})$ (Lemma 4.1) means the J-landscape at scale ratio x is identical to the landscape at x^{-1}. In RS, the small-x regime ($x \ll 1$, i.e., $\ln x < 0$) corresponds to quantum-mechanical action quantized by $\hbar$, while the large-x regime ($x \gg 1$) corresponds to gravitational curvature mediated by G. The reciprocal symmetry maps the quantum regime to the gravitational regime and vice versa.

For this mapping to be self-consistent in RS-native units—i.e., for the two regimes to have the same ledger structure—the couplings must satisfy:

$$G \cdot \hbar = 1. \tag{4}$$

This is the *quantum-gravitational duality.* Note that this is a structural argument: it assumes that G and $\hbar$ are the characteristic couplings of the two regimes related by $x \leftrightarrow x^{-1}$, and that the mapping preserves ledger structure. A fully rigorous derivation would require showing that the effective action at scale x and at scale x^{-1} are related by $G \cdot \hbar = 1$ through the RS dynamics; this derivation has not yet been completed and the duality is taken as a structural consequence of the reciprocal symmetry at this stage.

Since $\hbar = \varphi^{-5}$ (derived from the coherence energy $E_{\text{coh}} = \varphi^{-5}$):

$$G = \frac{1}{\hbar} = \frac{1}{\varphi^{-5}} = \varphi^5 \approx 11.090. \qquad \square \tag{5}$$

□

Corollary 4.3 (Planck Length Identity)**.** $\ell_P = \sqrt{\hbar G/c^3} = \sqrt{\varphi^{-5} \cdot \varphi^5/1} = 1$ *in RS-native units. The Planck scale equals the voxel scale.*

Remark 4.4 (Two derivation paths for G). The duality argument gives $G = \varphi^5$. An independent derivation via the recognition-wavelength extremum (Section 4.2) yields $G = \varphi^5/\pi$ when $\lambda_{\text{rec}} = \ell_0$ is used as the fundamental length. The two conventions differ by a factor of π: if $G = \varphi^5$ then $\lambda_{\text{rec}} = \ell_0/\sqrt{\pi} \neq \ell_0$; if $G = \varphi^5/\pi$ then $\lambda_{\text{rec}} = \ell_0$ but $\ell_P = \ell_0/\sqrt{\pi} \neq \ell_0$. In either case the ratio $\lambda_{\text{rec}}/\ell_P = 1/\sqrt{\pi} \approx 0.564$ is the same. This paper adopts $G = \varphi^5$ (duality convention). Resolving the π-factor between the two derivation paths is an open problem.

Remark 4.5. The duality $G \cdot \hbar = 1$ resolves the hierarchy problem. Gravity is "weak" not because G is anomalously small, but because $\hbar$ is anomalously large when measured in the same units. In RS-native units, $G = \varphi^5 \approx 11.09$ and $\hbar = \varphi^{-5} \approx 0.0902$—both are $O(1)$ on the φ-ladder. The apparent hierarchy between gravity and electromagnetism arises from expressing both in SI units, which are calibrated to the human-scale electronic regime.

4.2 The Einstein Coupling Constant

Definition 4.6 (Einstein Coupling)**.** The Einstein coupling constant is $\kappa = 8\pi G/c^4$.

Corollary 4.7 (κ Closed Form)**.** *In RS-native units ($c = 1$):*

$$\kappa = 8\pi\varphi^5 \approx 278.6. \tag{6}$$

Proof. Direct substitution of $G = \varphi^5$ and $c = 1$ into $\kappa = 8\pi G/c^4$. □

4.3 The Recognition Wavelength Identity

The recognition wavelength λ_{rec} provides an independent cross-check of the G derivation.

Proposition 4.8 (Recognition Wavelength Identity)**.**

$$\frac{c^3\,\lambda_{\text{rec}}^2}{\hbar\,G} = \frac{1}{\pi}. \tag{7}$$

Equivalently, $G = \pi\,c^3\,\lambda_{\text{rec}}^2/\hbar$.

Proof. The recognition wavelength is the length scale at which the J-cost gradient achieves its maximum curvature on the 3D ledger. Substituting $c = 1$, $\hbar = \varphi^{-5}$, and $G = \varphi^5$:

$$\lambda_{\text{rec}}^2 = \frac{\hbar\,G}{\pi\,c^3} = \frac{\varphi^{-5}\cdot\varphi^5}{\pi} = \frac{1}{\pi}. \tag{8}$$

Hence $\lambda_{\text{rec}} = 1/\sqrt{\pi} \approx 0.564$ in RS-native units. □

Remark 4.9. The recognition wavelength is not the voxel size: $\lambda_{\text{rec}} = \ell_0/\sqrt{\pi} \neq \ell_0$. It is a derived scale, smaller than the voxel size by a factor of $\sqrt{\pi}$. In SI units, $\lambda_{\text{rec}} \approx 9.12 \times 10^{-36}$ m.

5 SI Calibration

The calibration seam maps RS-native units to SI by identifying $\ell_0 = \ell_P$ and $\tau_0 = t_P$ (Corollary 4.3):

$$\ell_0 = \ell_P \approx 1.616 \times 10^{-35}\ \text{m}, \tag{9}$$

$$\tau_0 = t_P \approx 5.39 \times 10^{-44}\ \text{s}, \tag{10}$$

$$\hbar_{\text{SI}} = 1.055 \times 10^{-34}\ \text{J}\,\text{s}. \tag{11}$$

The SI value of G is recovered by dimensional analysis. In Planck units, $G_{\text{Pl}} = 1$ by definition. In RS-native units ($c = 1$, $\hbar = \varphi^{-5}$, $G = \varphi^5$) the Planck mass is $M_P = \sqrt{\hbar c/G} = \varphi^{-5}$, so 1 RS mass unit $= \varphi^5\,M_P$. Keeping $\ell_0 = \ell_P$ and $\tau_0 = t_P$:

$$G_{\text{SI}} = \varphi^5 \cdot \frac{\ell_P^3}{\varphi^5\,M_P\,t_P^2} = \frac{\ell_P^3}{M_P\,t_P^2} = G_{\text{Pl, SI}} \approx 6.674 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}. \tag{12}$$

This agrees with the CODATA value [2].

Remark 5.1. RS does not predict the SI number 6.674×10^{-11}—that depends on the human definition of meters, kilograms, and seconds. What RS predicts is the *dimensionless* identity $G \cdot \hbar = 1$ and the closed-form value $G = \varphi^5$ in natural units.

6 Why G Is Measured Poorly

In RS, G is determined by the Planck scale, while α is determined by the electronic scale. The Planck scale is $\sim 10^{17}$ times smaller, making it correspondingly harder to probe.

Different laboratory measurements of G disagree by $\sim$ 500 ppm [4], far exceeding quoted uncertainties. The RS framework suggests (but does not yet derive) a possible

structural explanation: the effective gravitational coupling at mesoscopic scales may receive small corrections from the φ-lattice structure.

$$G_{\text{eff}} = G \times (1 + \delta_{\text{lattice}}). \tag{13}$$

If $|\delta_{\text{lattice}}|$ depends on the relative orientation of the experimental apparatus with respect to the lattice, different setups (torsion balance [4] vs. atom interferometry [3]) would probe different aspects of this correction.

Remark 6.1. This lattice-correction hypothesis is *not* a derived result of RS; it is a speculative prediction motivated by the discrete structure. It would be falsified if future precision measurements converge to a single G value with no residual scatter.

7 The Gravity–QED Hierarchy

The ratio of electromagnetic to gravitational coupling for the electron is:

$$\frac{\alpha}{\alpha_{\text{grav}}} = \frac{e^2/(4\pi\epsilon_0)}{Gm_e^2} \approx 4.2 \times 10^{42}. \tag{14}$$

Proposition 7.1 (Hierarchy from the φ-Ladder — Structural Scaling)**.** *The ratio $\alpha/\alpha_{\text{grav}}$ is determined by the φ-rung separation Δr between the electron mass scale and the Planck scale:*

$$\frac{\alpha}{\alpha_{\text{grav}}} \sim \varphi^{2\Delta r}. \tag{15}$$

Proof. The gravitational "fine structure constant" for a particle of mass m at φ-rung r is $\alpha_{\text{grav}} = Gm^2/(\hbar c)$. Since $G \cdot \hbar = 1$, this becomes $\alpha_{\text{grav}} = m^2/c$. For $c = 1$, $\alpha_{\text{grav}} = m^2$. The mass at rung r is $m \propto \varphi^r$, so $\alpha_{\text{grav}} \propto \varphi^{2r}$. The electromagnetic coupling α is at the electron rung r_e, while the gravitational coupling becomes $O(1)$ at the Planck rung r_{Pl}. The ratio scales as $\varphi^{2(r_{\text{Pl}} - r_e)}$. □

Remark 7.2. The hierarchy is not a "problem" in RS; it reflects the geometric distance on the φ-ladder between the relevant scales. There is no fine-tuning: the rung separation is a structural consequence of the ledger, not an accidental cancellation.

8 Running of G

The φ-ladder structure suggests that G may run with energy scale, analogous to the running of gauge couplings. The discrete nature of the ladder introduces a natural running exponent:

$$\beta_G = -\frac{\varphi - 1}{\varphi^5} = -\frac{1/\varphi}{\varphi^5} = -\varphi^{-6} \approx -0.056. \tag{16}$$

Remark 8.1. This running is unmeasurable at macroscopic scales: deviations from the constant-G prediction are suppressed by $(E/E_{\text{Pl}})^{|\beta_G|}$. The prediction becomes testable only in the quantum-gravity regime ($E \sim E_{\text{Pl}}$), where $G(E) \to 0$ (consistent with the asymptotic safety scenario).

The running exponent $\beta_G = -\varphi^{-6}$ is a *hypothesis* motivated by the φ-ladder, not a rigorously derived result. It would be falsified by any measurement of G at high energies inconsistent with this prediction.

9 Predictions

1. $G \cdot \hbar = 1$ (dimensionless, in RS-native units). This is the primary falsifiable prediction: any experiment probing the Planck scale that contradicts this identity would falsify the RS reciprocal-symmetry argument.

2. $G = \varphi^5$ (RS-native). The closed-form value has zero free parameters.

3. **No fifth force.** G is universal—all matter couples to gravity with the same G because all mass arises from the same J-cost function. Any experiment detecting a composition-dependent gravitational coupling would falsify RS.

4. $\ell_P = \ell_0$**.** The Planck length equals the voxel size. There is no "trans-Planckian" regime: the Planck scale is the lattice scale.

5. **Running of** G (hypothesis). G runs with exponent $\beta_G \approx -0.056$, significant only near the Planck scale.

10 Conclusion

Newton's gravitational constant is derived from the RS forcing chain via the quantum-gravitational duality:

$$G = \frac{1}{\hbar} = \varphi^5 \approx 11.09 \quad \text{(RS-native units)}. \tag{17}$$

The duality $G \cdot \hbar = 1$ follows from the reciprocal symmetry $J(x) = J(x^{-1})$ of the cost functional. The Einstein coupling is $\kappa = 8\pi\varphi^5$. The hierarchy between gravity and electromagnetism is explained by φ-ladder geometry. G is not a free parameter but a consequence of the coherence energy—determined entirely by the cost functional $J(x)$ and the spatial dimension $D = 3$.

References

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